

Directions: Use the following situation to answer both the multiple choice questions that follow.

A surprising number of young adults (ages 19 to 25) still live in their parents' homes. A random sample by the National Institutes of Health included 2252 men and 2629 women in this age group. The survey found that 986 of the men and 923 of the women lived with their parents. Let p_W = the proportion of women who still live at home, and let p_M = the proportion of men who still live at home.

_____ 1) Which of the following statements about the conditions necessary for constructing a confidence interval for the difference in population proportions is INCORRECT?

- (A) Because it is a random sample, we know that the center for the sampling distribution for the difference in sample proportions is unbiased and centered at $p_M - p_W$
- (B) Because we are sampling, we need to check that the sample size is less than 10% of the population size in order to correctly sample without replacement from our population.
- (C) The large counts condition needs to be checked and satisfied for both samples in order to conclude that the sampling distribution for the difference in sample proportions is approximately normal.
- (D) Because our samples are so large, the CLT tells us the sampling distribution will be approximately normal, so there is no need to check the large counts condition.
- (E) We would use the standard error to approximate the sampling distribution standard deviation for the difference in sample proportions.

_____ 2) What is the 98% confidence interval for the difference in the true proportions of men and women ages 19 to 25 ($p_M - p_W$) who live in their parents' homes?

- (A) $(0.438 - 0.351) \pm 2.326 \sqrt{\frac{0.438(0.562)}{2252} + \frac{0.351(0.649)}{2629}}$
- (B) $(0.438 - 0.351) \pm 2.326 \sqrt{(0.391)(0.609) \left(\frac{1}{2252} + \frac{1}{2629} \right)}$
- (C) $(0.438 - 0.351) \pm 1.96 \sqrt{\frac{0.438(0.562)}{2252} + \frac{0.351(0.649)}{2629}}$
- (D) $(0.438 - 0.351) \pm 1.96 \sqrt{(0.391)(0.609) \left(\frac{1}{2252} + \frac{1}{2629} \right)}$
- (E) $(0.438 - 0.351) \pm 2.326 \sqrt{\sqrt{\frac{0.438(0.562)}{2252}} + \sqrt{\frac{0.351(0.649)}{2629}}}$

_____ 3) What is the correct interpretation of the confidence interval from question 2?

- (A) We are 98% confidence that the difference in sample proportions of men and women ages 19 to 25 who live in their parents' homes is between 0.054 and 0.119.
- (B) We are 98% confidence that the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes is between 0.054 and 0.119.
- (C) We are 98% confidence that more men, who are 19 to 25 years old, live at home than women who are 19 to 25 years old.
- (D) If we were to repeat this survey again, we are 98% confidence that the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes will be between 0.054 and 0.119.
- (E) If we were to repeat this survey many times, 98% of the confidence intervals will correctly capture the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes

_____ 4) What is the correct interpretation of the confidence level in context of this problem?

- (A) If we were to repeat this survey 100 times, 98 of the results confidence intervals will report that the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes will be between 0.054 and 0.119.
- (B) We are 98% confidence that the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes is between 0.054 and 0.119.
- (C) There is a 98% chance that the interval we created captures the true population parameter of the difference in the proportions of men and women ages 19 to 25 who live in their parents' homes.
- (D) If we were to repeat this survey again, we are 98% confidence that the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes will be between 0.054 and 0.119.
- (E) If we were to repeat this survey many times, 98% of the confidence intervals will correctly capture the true difference in the proportions of men and women ages 19 to 25 who live in their parents' homes

_____ 5) With all other aspects remaining the same, what would happen to our confidence interval if we change the interval from a 98% confidence interval to a 90% confidence interval?

- (A) The interval will get narrower, since the critical value increases.
- (B) The interval will get narrower, since the critical value decreases.
- (C) The interval will get wider, since the critical value increases.
- (D) The interval will get wider, since the critical value decreases.
- (E) The interval will remain the same width, but the point estimate will shift to a lower value.

_____ 6) If we changed our difference from $p_M - p_W$ to $p_W - p_M$ but all other elements remain the same, what will happen?

- I. The point estimate will change from positive to negative
- II. The margin of error will remain the same
- III. The new confidence interval will contain 0

- (A) I only (B) I and II only (C) II and III only (D) I and III only (E) I, II, III